

Section A

Question 1

This is false negatives

The correct condition should include patients with HbA1c equal to 6.5%, but the programmer used a strict greater-than condition. This means patients with a value of exactly 6.5% were excluded, even though they meet the clinical definition of diabetes.

So in this case, the patients do have the condition, but the system classified them as healthy. That is why they are considered false negatives. As I understand,

$$\text{False Negative: Actual} = 1, \text{ Predicted} = 0$$

Question 2

It would have to be sensitivity

Here the focus is on patients who truly have chronic diabetes, and how many of them were correctly identified using the insulin data. So we are starting from the actual positive group and checking how many were picked up correctly.

So this is what sensitivity measures. It looks at how well the method captures true cases without missing them.

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

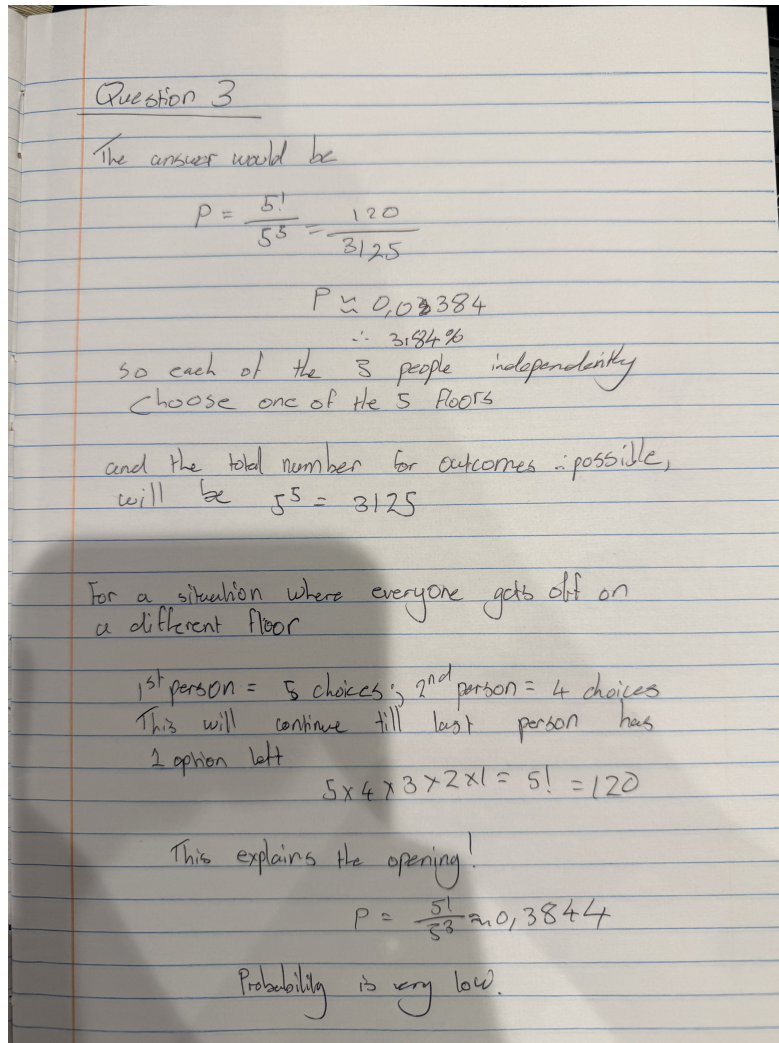
A higher sensitivity means fewer missed cases, which is especially important in a health-care setting where missing a diagnosis can have serious consequences.

Question 3

the probability is:

$$P = \frac{120}{3125} \approx 0.0384$$

For completeness, my full working is shown below:



Question 4

This is a linear regression example, where the red line is used to describe the relationship between the variables.

For this line to be a reasonable representation of the data, let go through the following assumptions,

Now firstly, we assume that the relationship between the variables can be approximated by a straight line. If the true relationship is curved or more complex, then a linear model would not work well.

We also assume that the observations are independent, this means that one data point does not directly influence another.

Another important assumption is that the spread of the errors remains roughly constant across the range of the data. In other words, the variability around the line should not increase or decrease as the values change.

Finally, the residuals are assumed to be approximately normally distributed:

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

Here, ϵ represents the difference between the actual value and the value predicted by the model. This assumption means that these errors are centred around zero, so the model is not consistently over- or under-predicting. The term σ^2 describes how spread out these errors are.

So in when you implement this in practice, this means most points lie reasonably close to the line, while larger deviations occur less frequently. This assumption becomes important when making statistical inferences, such as constructing confidence intervals or testing relationships.

Question 5

Looking at the diagram, most of the bullet holes are concentrated on the wings and the outer sections of the aircraft. At first glance at this, it might seem reasonable to reinforce those areas, since they appear to be hit the most.

However, this conclusion is misleading because the data only comes from planes that were able to return. This means we are only seeing damage that the aircraft could survive.

The areas with many bullet holes are therefore not the most critical, since planes were still able to fly back despite being hit there. The more important insight comes from the areas with very few or no bullet holes, such as around the engine and central body. If those parts are hit, the aircraft likely does not return, which is why those hits are missing from the data.

Because of this, armour should be added to the areas with the fewest bullet holes, as these represent the most vulnerable parts of the plane.

In statistical terms, this is an example of survivorship bias. The data reflects $P(\text{damage} \mid \text{survived})$ rather than the true distribution of damage across all aircraft, which can lead to incorrect conclusions if not interpreted carefully.